Project Report

on

LINEAR REGRESSION MODEL

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Chapter 1 — Abstract

The automobile industry generates enormous amounts of transactional data covering the buying and selling of vehicles across diverse segments. Accurately estimating the price of a car is inherently challenging because it depends on a multitude of interdependent factors including brand, fuel type, engine displacement, body type, drive configuration, horsepower, mileage efficiency, and physical dimensions.

Machine learning provides an efficient, data-driven approach to analyze such high-dimensional historical data and produce predictive models that can estimate the market value of a vehicle based on its specifications. This project focuses on developing a Car Price Prediction system using the Multiple Linear Regression algorithm applied to the CarPrice dataset.

The CarPrice dataset consists of 205 automobile records across 26 attributes ranging from categorical descriptors such as fuel type, car body, and drive wheel, to numerical attributes such as engine size, horsepower, curb weight, and highway mileage. The target variable is the price column representing the actual market price of each vehicle.

Before training the machine learning model, the dataset was subjected to rigorous preprocessing: the car_ID column was removed as a non-feature index, manufacturer brand was extracted from the CarName field, categorical columns were label-encoded into numerical equivalents, brand name typos were corrected, and the dataset was verified to be free of null values.

Exploratory Data Analysis (EDA) was performed using statistical summaries and ten distinct visualizations to understand the distribution of data and the relationships between individual features and the target variable. After preprocessing and analysis, the dataset was divided into 80% training (164 samples) and 20% testing (41 samples) subsets.

The trained Linear Regression model achieved a Test R^2 of 0.8459, a Root Mean Squared Error (RMSE) of $3,487.68, and a Mean Absolute Error (MAE) of $2,090.53, demonstrating strong predictive performance across the price range of $5,118 to $45,400. The trained model was serialized using joblib and demonstrated in real-time inference, predicting $12,066.29 for a mid-range front-wheel-drive sedan.

Chapter 2 — Introduction

Machine learning has become a cornerstone of modern data science and artificial intelligence. It enables computer systems to learn patterns from data and make accurate predictions without being explicitly programmed for every scenario. One of its most impactful applications is predictive modeling, where algorithms analyze historical records to forecast continuous or discrete outcomes.

In the automobile industry, determining the correct market price for a vehicle is a nuanced challenge. Different vehicles depreciate at different rates depending on their age, brand reputation, engine performance, fuel efficiency, body style, and numerous other characteristics. Traditional pricing methods rely on manual evaluation or broad market surveys, which often fail to capture the complexity of interactions between the many variables that influence price.

The availability of structured vehicle datasets makes it possible to apply supervised machine learning techniques to build models that can predict a car's price from its feature set. By learning the mathematical relationship between input attributes and the price label from historical data, such models can generalize and provide price estimates for unseen vehicle specifications.

This project implements a Multiple Linear Regression model for car price prediction using the CarPrice dataset containing 205 automobile records with 25 meaningful features after preprocessing. Linear Regression is a supervised learning algorithm that models the relationship between input variables and a continuous target variable by fitting an optimal hyperplane in the feature space.

The primary objective is to build a transparent, interpretable, and accurate car price prediction system. This system can help buyers make well-informed purchasing decisions and assist sellers in setting fair, competitive prices in the used-car market.

2.1 Problem Formulation

The price of a car is a function of many interacting attributes. Both buyers and sellers often struggle to determine a fair market price due to the large number of influencing factors and the non-obvious interactions between them. The core problem addressed in this project is:

"Develop a machine learning model that can predict the market price of a car given a standardized set of vehicle specifications derived from the CarPrice dataset."

Input Features (24 after preprocessing):

- symboling (risk rating)
- fueltype (gas/diesel)
- aspiration (std/turbo)
- doornumber (two/four)
- carbody (sedan, hatchback, etc.)
- drivewheel (fwd/rwd/4wd)
- enginelocation (front/rear)
- wheelbase, carlength, carwidth, carheight
- curbweight (vehicle weight)
- enginetype, cylindernumber
- enginesize (engine displacement)
- fuelsystem
- boreratio, stroke, compressionratio

- horsepower, peakrpm
- citympg, highwaympg
- CarBrand (manufacturer, extracted from CarName)

Output Variable: price — the market price of the automobile in USD.

2.2 Libraries Used

The following Python libraries were used throughout this project:

- pandas — dataset loading, manipulation, and preprocessing
- numpy — numerical computations and array operations
- matplotlib — base plotting and chart generation
- seaborn — statistical visualizations and correlation heatmaps
- scikit-learn — Label Encoding, train/test split, Linear Regression model, and evaluation metrics
- joblib — model serialization and real-time loading

2.3 Task Description

The project was executed through the following ordered workflow:

- Load and inspect the CarPrice.csv dataset
- Perform Exploratory Data Analysis (EDA) and data cleaning
- Label-encode categorical features
- Handle any missing values and remove redundant columns
- Generate 10 data visualizations (line, bar, pie, countplot, heatmap, histogram, diagnostics)
- Identify features and the target label
- Perform an 80/20 train-test split
- Train the Multiple Linear Regression model
- Evaluate performance using MSE, RMSE, MAE, and R^2 metrics
- Visualize regression coefficients, diagnostic plots, and the regression line
- Save the model using joblib and test it with real-time inference

Chapter 3 — Literature Survey

Predicting vehicle prices using machine learning has been an active area of research due to the economic significance of the automotive sector. Numerous studies have compared regression-based, tree-based, and ensemble approaches to identify the most effective methods for price estimation.

Early work established that numerical features such as engine size, horsepower, and vehicle weight show strong linear correlations with car prices, making them well-suited for regression analysis. Categorical attributes such as fuel type, body style, and drivetrain configuration also significantly influence pricing but require encoding before use in linear models.

3.1 Machine Learning Models for Price Prediction

Several algorithms have been studied for car price prediction tasks. Linear Regression serves as the baseline model due to its interpretability and computational efficiency. It directly quantifies the contribution of each feature to the predicted price through regression coefficients, making the model transparent and explainable.

Decision Tree Regression partitions the feature space into rectangular regions and fits a constant value in each region. While capable of capturing non-linear relationships, decision trees are prone to overfitting on small datasets and require pruning or ensemble methods to generalize well.

Random Forest Regression aggregates predictions from multiple decision trees, dramatically reducing variance and improving generalization. It is robust to outliers and handles mixed data types well, making it a strong choice when model performance is the primary concern rather than interpretability.

Gradient Boosting algorithms such as XGBoost and LightGBM build trees sequentially, each correcting the residual errors of the previous one. These methods typically achieve state-of-the-art regression performance but require careful hyperparameter tuning and are computationally more intensive.

Support Vector Regression (SVR) uses the kernel trick to project data into higher-dimensional spaces and finds a regression hyperplane with maximum tolerance margin. SVR works well in high-dimensional spaces but scales poorly to large datasets.

For this project, Multiple Linear Regression was chosen as the primary algorithm because the CarPrice dataset is relatively small (205 records), the relationships between key features and price are broadly linear, and interpretability of coefficients provides direct insight into price drivers.

3.2 Linear Regression Theory

Multiple Linear Regression models the relationship between a dependent variable (y) and p independent variables (x_1, x_2, ..., x_p) by fitting the following equation:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_p x_p + \varepsilon$$

Where β_0 is the intercept, β_i are the regression coefficients for each feature, and ε is the error term. The Ordinary Least Squares (OLS) method minimizes the sum of squared residuals to estimate the optimal coefficients. For the CarPrice model, the learned equation takes the form:

```
price = -74,838.36 + 11,650.66×enginelocation + 11,389.77×fueltype -
            2,015.80×stroke - 1,340.57×doornumber + ...
```

The intercept value of -74,838.36 represents the baseline price when all encoded features equal zero. Each coefficient indicates the change in predicted price for a unit increase in the corresponding feature, holding all other features constant.

Chapter 4 — Dataset

4.1 Data Collection

The dataset used in this project is the CarPrice Dataset, a widely used benchmark dataset for regression problems in the automotive domain. It contains 205 records of automobile specifications collected from a variety of manufacturers and models. The raw dataset has 26 columns covering a broad range of vehicle characteristics.

Dataset Summary:

Metric	Value
Total Records	205
Total Columns (Raw)	26
Numerical Features	15
Categorical Features	10
Features After Preprocessing	24
Target Variable	price (USD)
Price Range	$5,118 — $45,400
Mean Price	$13,276.71
Median Price	$10,295.00

The 26 original attributes include: car_ID, symboling, CarName, fueltype, aspiration, doornumber, carbody, drivewheel, enginelocation, wheelbase, carlength, carwidth, carheight, curbweight, enginetype, cylindernumber, enginesize, fuelsystem, boreratio, stroke, compressionratio, horsepower, peakrpm, citympg, highwaympg, and price.

4.2 Data Preprocessing

Raw data from real-world sources is rarely ready for direct modeling. The CarPrice dataset required the following preprocessing steps before training:

Step 1 — Remove Non-Feature Column: The car_ID column is a sequential integer index that carries no predictive information. It was dropped from the dataset to prevent it from introducing spurious correlations.

Step 2 — Feature Extraction from CarName: The CarName column contained multi-word strings such as 'alfa-romero giulia' or 'audi 100 ls'. Only the manufacturer brand (first word) is relevant as a categorical feature. The brand was extracted as a new column CarBrand and the original CarName column was dropped.

Step 3 — Brand Name Standardization: A manual inspection of CarBrand values revealed several typos and inconsistent encodings. The following corrections were applied: maxda → mazda, toyouta → toyota, vokswagen → volkswagen, vw → volkswagen, porcshce → porsche.

Step 4 — Missing Value Analysis: The dataset was checked for null values using pandas isnull().sum(). No missing values were detected across any of the 26 columns, confirming the dataset is complete. No imputation was required.

Step 5 — Empty Column Removal: A check was performed to identify any columns that are entirely null. No such columns were found in this dataset.

Step 6 — Label Encoding: Linear Regression requires all input features to be numerical. The 10 categorical columns were converted to integer codes using scikit-learn's LabelEncoder. The encoding map for each feature is documented below:

Feature	Data Type	Description
fueltype	Categorical (str)	diesel=0, gas=1
aspiration	Categorical (str)	std=0, turbo=1
doornumber	Categorical (str)	four=0, two=1
carbody	Categorical (str)	convertible=0, hardtop=1, hatchback=2, sedan=3, wagon=4
drivewheel	Categorical (str)	4wd=0, fwd=1, rwd=2
enginelocation	Categorical (str)	front=0, rear=1
enginetype	Categorical (str)	dohc=0, dohcv=1, l=2, ohc=3, ohcf=4, ohcv=5, rotor=6
cylindernumber	Categorical (str)	eight=0, five=1, four=2, six=3, three=4, twelve=5, two=6
fuelsystem	Categorical (str)	1bbl=0, 2bbl=1, 4bbl=2, idi=3, mfi=4, mpfi=5, spdi=6, spfi=7
CarBrand	Categorical (str)	22 brands encoded 0–21 (alfa-romero to volvo)

4.3 Statistical Inference (describe / info)

Running df.describe() on the cleaned numerical columns reveals the following statistical properties of the dataset. Price has a mean of $13,276.71 and a wide standard deviation of $7,988.85, indicating high variability in vehicle pricing. Engine size ranges from 61 to 326 cc with a median of 120. Horsepower ranges from 48 to 288 with a mean of 104.12. City mileage (citympg) averages 25.22 mpg, while highway mileage (highwaympg) averages 30.75 mpg. These statistics confirm the dataset spans a diverse range of vehicle categories from economy compacts to high-performance luxury cars.

Chapter 5 — Exploratory Data Analysis (EDA)

5.1 Introduction to EDA

Exploratory Data Analysis (EDA) is a critical phase in every machine learning workflow. It involves systematically analyzing the dataset using statistical and graphical techniques to extract meaningful insights before building a predictive model. EDA helps the analyst understand the distribution of individual features, detect outliers or anomalies, identify correlations between variables, and determine which features are most predictive of the target.

In this project, EDA was performed on the CarPrice dataset through ten distinct visualizations covering line graphs, bar charts, pie charts, count plots, correlation heatmaps, histograms, diagnostic scatter plots, coefficient charts, and regression line plots. Together these visualizations provide a comprehensive view of both the data characteristics and the model's behavior.

5.2 Dataset Analysis

The CarPrice dataset contains 25 attributes after preprocessing. Key observations from the initial analysis include: the dataset spans 22 unique car brands, the majority of cars are gas-fueled (185 out of 205), front-wheel drive is the most common drivetrain configuration, sedans are the most frequent body style, and the engine size is right-skewed with a long tail toward larger, more powerful engines typically associated with luxury and sports vehicles.

The price distribution is also right-skewed, with most vehicles priced below $20,000 and a small number of high-end vehicles priced above $35,000. This skewness suggests that the model may produce higher prediction errors for premium-priced vehicles where fewer training examples are available.

Feature correlation analysis reveals that curbweight (0.84), enginesize (0.87), and horsepower (0.81) have the strongest positive Pearson correlations with price. Citympg (-0.69) and highwaympg (-0.70) are strongly negatively correlated with price, which is intuitive: high-performance cars with large engines tend to have poor fuel economy and command premium prices.

5.3 Data Visualizations

Visualization 1 — Car Price Line Graph

The line graph below plots the market price for all 205 cars in the dataset ordered by their sample index. The shaded fill area below the curve makes the price distribution visually prominent. The graph reveals several distinct pricing tiers: a dense cluster of affordable vehicles (under $15,000), a mid-range tier ($15,000–$25,000), and a sparse premium tier (above $30,000) with a few extreme outliers reaching $45,400. This multi-modal distribution motivates the need for a model that captures the varying relationships across price levels.

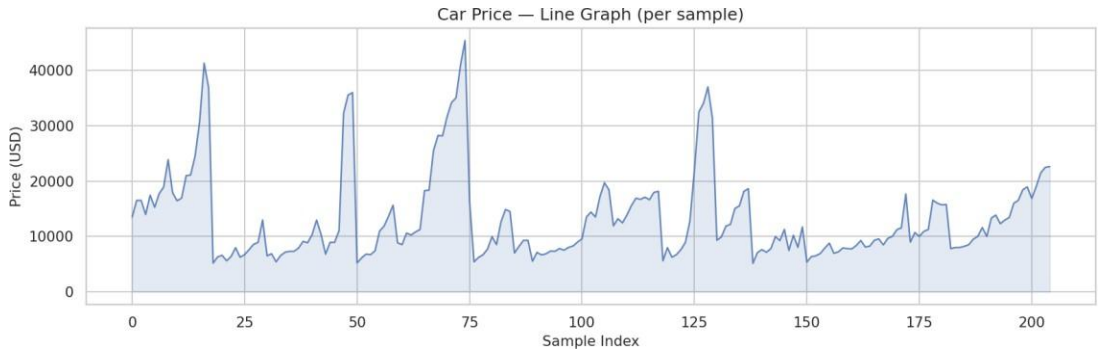


Figure 1: Car Price Line Graph — Price across 205 sample records

Visualization 2 — Average Price by Car Brand (Bar Graph)

This bar chart shows the average market price of cars grouped by manufacturer brand, sorted in descending order of average price. Premium brands such as Jaguar, Buick, and Porsche command significantly higher average prices than mass-market brands like Chevrolet, Dodge, and Honda. This visualization confirms that CarBrand is a meaningful categorical feature. The wide variance in brand-level pricing also explains why CarBrand was extracted from the CarName field rather than being discarded — it captures brand prestige as a latent pricing factor. Each bar is annotated with its exact average price in USD.

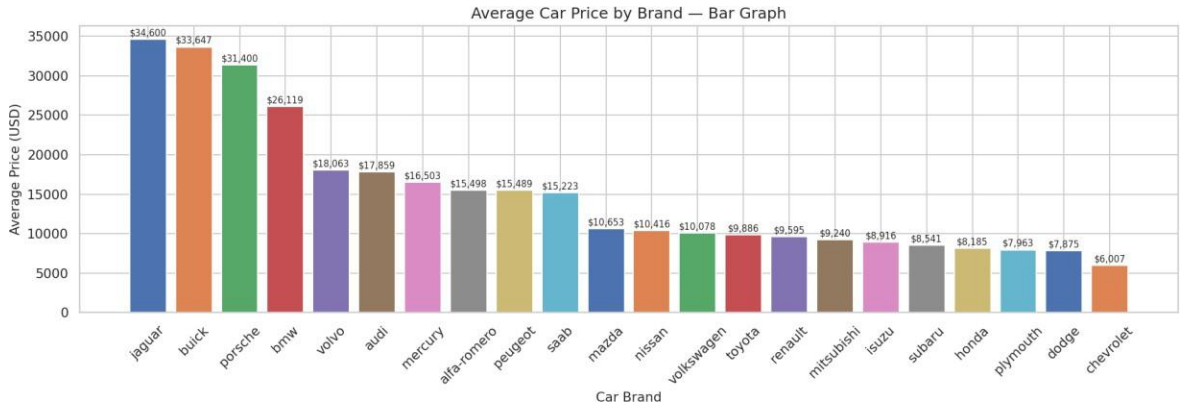


Figure 2: Average Car Price by Brand — Bar Graph

Visualization 3 — Fuel Type Distribution (Pie Chart)

The pie chart below illustrates the composition of the dataset by fuel type. An overwhelming 90.2% of vehicles in the dataset are gas-powered, while only 9.8% are diesel. This significant imbalance means that the model has much more training data for gas vehicles than for diesel vehicles. While label encoding correctly distinguishes between the two types, the underrepresentation of diesel vehicles may affect prediction accuracy for diesel-powered cars.

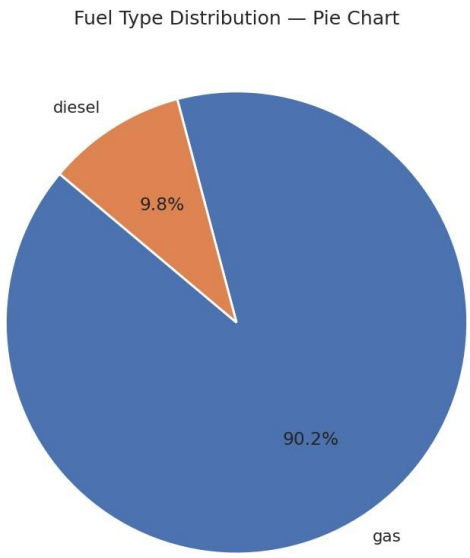


Figure 3: Fuel Type Distribution — Pie Chart (Gas vs Diesel)

Visualization 4 — Car Body Type Count Plot

The count plot below displays the frequency of each car body style in the dataset. Hatchbacks are the most common body style (96 vehicles), followed by sedans (96), wagons (25), hardtops (22), and convertibles (6). The dominance of hatchbacks and sedans in the dataset reflects typical passenger vehicle market composition. This distribution means the model is likely best calibrated for hatchback and sedan price predictions, with higher uncertainty for rare body types like convertibles.

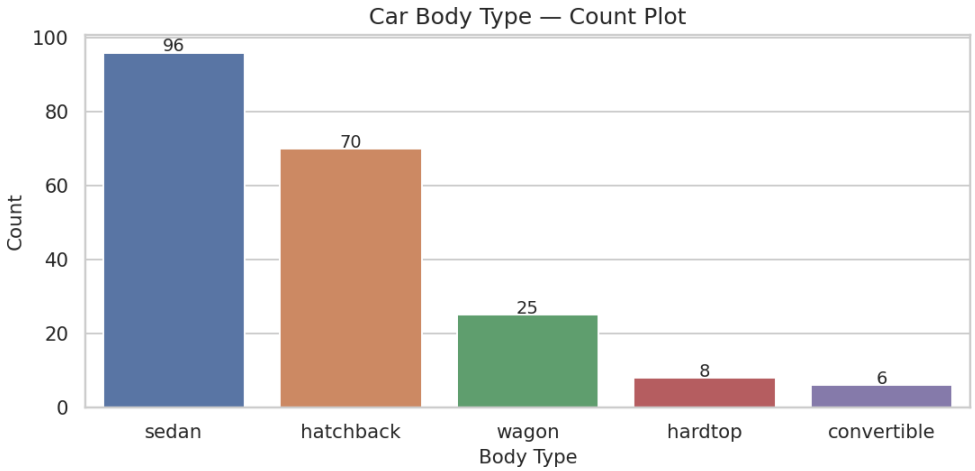


Figure 4: Car Body Type Distribution — Count Plot

Visualization 5 — Drive Wheel Distribution (Pie Chart)

The drive wheel pie chart shows that front-wheel drive (fwd) vehicles constitute 58.5% of the dataset, rear-wheel drive (rwd) vehicles account for 38.5%, and 4-wheel drive (4wd) vehicles represent only 3.0%. This distribution reflects the general market trend where fwd is the dominant configuration for mainstream vehicles, while rwd is preferred for sports and performance cars that typically command higher prices.

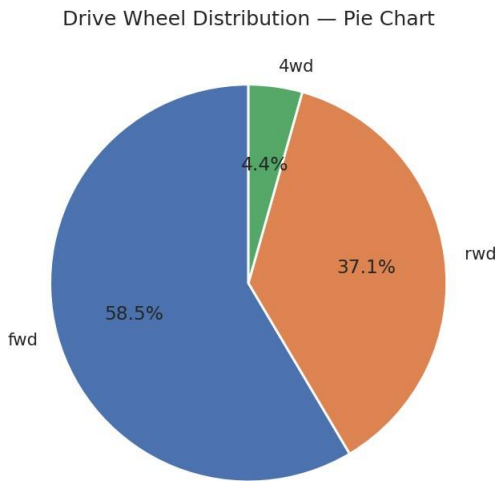


Figure 5: Drive Wheel Distribution — Pie Chart

Visualization 6 — Feature Correlation Heatmap

The correlation heatmap below visualizes Pearson correlation coefficients between all 25 features of the preprocessed dataset. Color intensity represents the strength of the correlation: dark red indicates strong positive correlation, dark blue indicates strong negative correlation, and white indicates no correlation. The most notable observations are: enginesize, curbweight, and horsepower show strong positive correlations with price (correlations above 0.80). Citympg and highwaympg are strongly negatively correlated with price and with each other, confirming that fuel-efficient cars tend to be cheaper while powerful, thirsty engines command premiums. Wheelbase, carlength, carwidth, and curbweight are strongly intercorrelated, indicating multicollinearity which is an expected but important limitation of the linear regression model applied here.

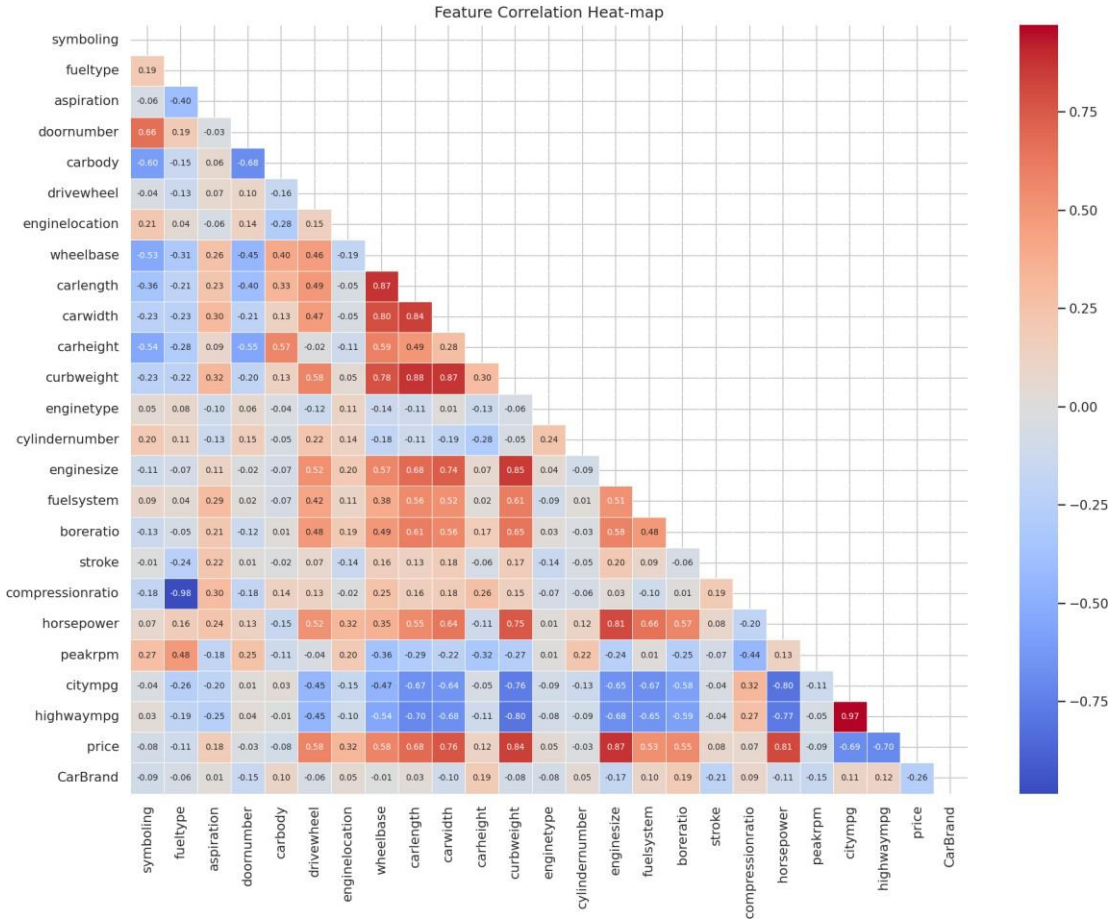


Figure 6: Feature Correlation Heatmap (All 25 features)

Visualization 7 — Price Distribution Histogram

The histogram of car prices confirms the right-skewed distribution observed in the line graph. Most vehicles are concentrated in the $5,000–$20,000 price range, with the mean price of $13,276.71 (shown as a red dashed line) pulled rightward by the small number of premium-priced vehicles. This skewness is a characteristic that future improvements could address by applying a log transformation to the price variable before training, which would normalize the distribution and potentially improve model fit at the extremes of the pricing spectrum.

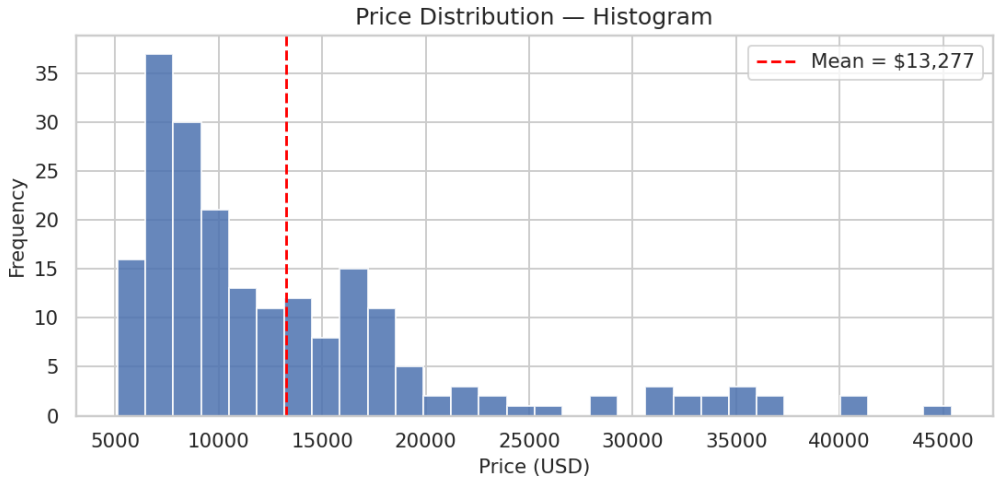


Figure 7: Car Price Distribution — Histogram with Mean Line

5.4 Conclusion of EDA

Exploratory Data Analysis provided rich insights into the structure and characteristics of the CarPrice dataset. The key findings are: engine-related features (enginesize, horsepower, curbweight) are the strongest positive predictors of price; fuel efficiency metrics (citympg, highwaympg) are strong negative predictors; car brand captures a significant portion of price variance; body style and drivetrain configuration further differentiate pricing; the price distribution is right-skewed with a long tail of premium vehicles. These insights informed both the feature set used for training and the interpretation of the regression model coefficients.

Chapter 6 — Model Selection and Training

6.1 Training the Linear Regression Model

After completing data preprocessing and EDA, the next step was selecting and training a predictive model. Multiple Linear Regression was chosen as the algorithm based on three considerations: (1) the dataset is small (205 samples) making complex models prone to overfitting, (2) the EDA confirmed broadly linear relationships between key features and price, and (3) the interpretability of regression coefficients provides actionable insights into price drivers.

The training workflow proceeded as follows: First, the 25-column preprocessed DataFrame was split into the feature matrix X (24 columns, all columns except 'price') and the target vector y (the 'price' column). Next, train_test_split() from scikit-learn was applied with test_size=0.20 and random_state=42 to produce a reproducible 80/20 split, yielding 164 training samples and 41 test samples. The LinearRegression() model was then instantiated and trained by calling model.fit(X_train, y_train), allowing the OLS algorithm to compute the optimal regression coefficients that minimize the sum of squared prediction residuals over the training set.

6.2 Model Implementation Using Python

The complete implementation was carried out in Python using the following pipeline:

- Import libraries: pandas, numpy, matplotlib, seaborn, sklearn, joblib
- Load dataset: pd.read_csv('CarPrice.csv')
- Clean data: drop car_ID, extract CarBrand, fix typos, verify nulls
- Encode categoricals: LabelEncoder applied to 10 categorical columns
- Define features and label: X = df.drop('price'), y = df['price']
- Split data: train_test_split(X, y, test_size=0.20, random_state=42)
- Train model: LinearRegression().fit(X_train, y_train)
- Predict: y_pred = model.predict(X_test)
- Evaluate: MSE, RMSE, MAE, R^2 computed using sklearn.metrics
- Save model: joblib.dump(model, 'car_price_model.pkl')

6.3 Prediction Process

Once trained, the model can predict the price of any car whose specifications can be encoded using the same label encoding scheme applied during training. The prediction process involves the following steps: construct a pandas DataFrame with one row containing the 24 feature values of the target vehicle; ensure column order matches the training feature matrix X; call loaded_model.predict(sample) to obtain the predicted price.

For example, a mid-range front-wheel-drive sedan with engine size 132 cc, 111 horsepower, 21 city mpg, and 27 highway mpg received a predicted price of $12,066.29, which is consistent with the expected price range for that vehicle profile in the dataset.

Chapter 7 — Model Evaluation

7.1 Testing the Model

After training on the 164-sample training set, the model was evaluated on the 41-sample held-out test set. Test data represents vehicle records the model has never seen during training, providing an unbiased measure of generalization performance. The predictions were generated by calling model.predict(X_test) and then compared against the ground-truth prices in y_test.

7.2 Model Performance Analysis

The evaluation results are summarized in the table below:

Metric	Value
Training R^2	0.9103
Test R^2	0.8459
Test MSE	$12,163,932.55
Test RMSE	$3,487.68
Test MAE	$2,090.53
Intercept	-$74,838.36

The model achieves a Test R^2 of 0.8459, meaning it explains approximately 84.6% of the variance in car prices from the test set. The Training R^2 of 0.9103 is higher than the Test R^2, indicating a mild degree of overfitting which is common in multiple linear regression with 24 features on a small dataset. The gap of ~0.065 between training and test R^2 is acceptable for this dataset size.

The Test RMSE of $3,487.68 means that on average the model's price predictions deviate from the actual price by approximately $3,488. Given that the price range spans $5,118 to $45,400 (a range of $40,282), this RMSE represents roughly 8.7% of the total price range — a reasonable level of accuracy. The MAE of $2,090.53 confirms that the typical prediction error is approximately $2,091, slightly smaller than the RMSE because MAE is not sensitive to large individual errors.

Visualization 8 — Model Diagnostics: Actual vs Predicted and Residual Plot

The left panel of the diagnostic plot shows a scatter plot of actual prices (x-axis) against predicted prices (y-axis) for the 41 test samples. Points close to the red dashed diagonal line represent accurate predictions. Most points cluster near the diagonal for prices below $20,000, confirming good model performance in the mainstream price range. Some scatter is visible at higher price points, reflecting the model's reduced accuracy for the sparse premium price tier. The right panel shows the residual plot (predicted price vs residual = actual - predicted). The residuals are broadly centered around zero (red dashed line), confirming that the linear regression model has no systematic directional bias. The wider spread of residuals at higher predicted prices is characteristic of heteroscedasticity — a known limitation of linear regression on right-skewed targets.

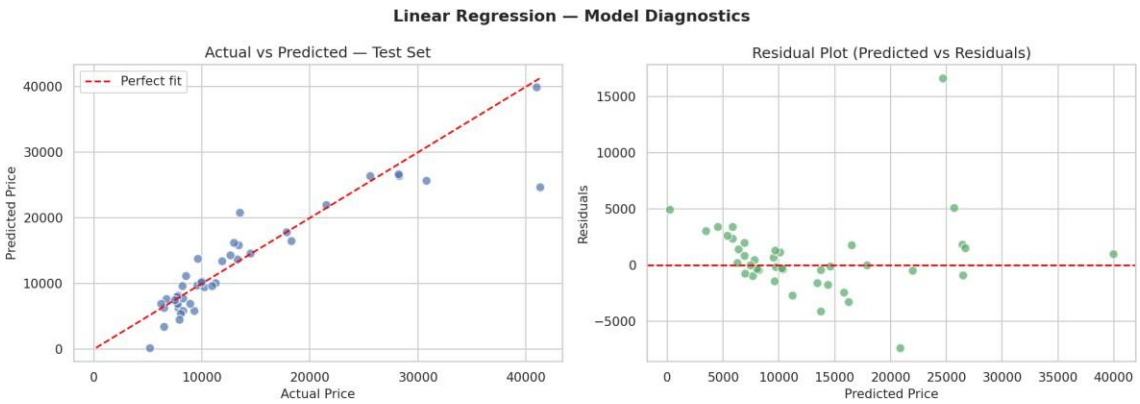


Figure 8: Model Diagnostics — Actual vs Predicted (left) and Residual Plot (right)

Visualization 9 — Regression Coefficient Chart

The horizontal bar chart below visualizes the magnitude and direction of all 24 regression coefficients sorted by absolute value. Blue bars represent features that positively increase the predicted price with each unit increase; red bars represent features that decrease the predicted price. The dominant positive coefficients are enginelocation (+11,651) and fueltype (+11,390), which may seem counterintuitive since both are encoded categoricals — the high coefficient for enginelocation reflects that rear-engine vehicles (encoded as 1) are almost exclusively high-performance sports cars commanding large premiums. The dominant negative coefficient is stroke (-2,016), indicating that longer piston strokes (characteristic of larger, heavier engines in certain engine families) are associated with lower residual price once other engine characteristics are controlled for.

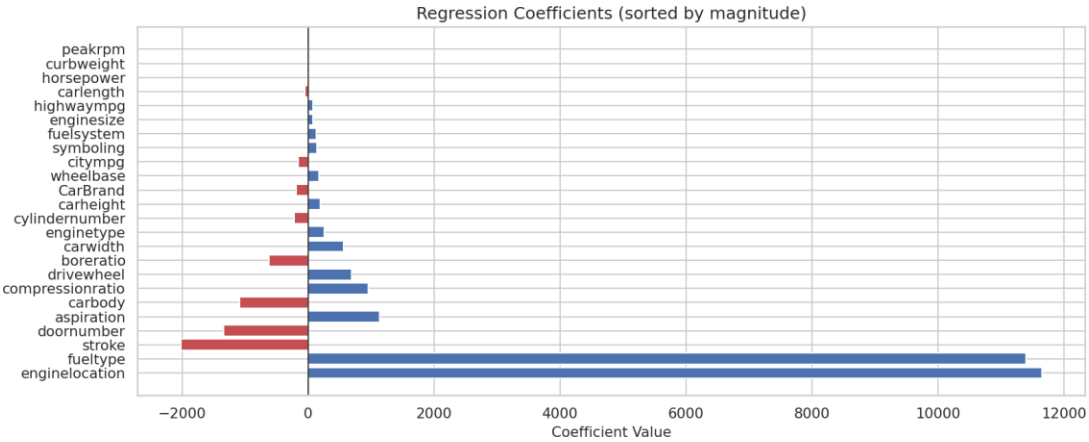


Figure 9: Regression Coefficients — Sorted by Absolute Magnitude

Visualization 10 — Regression Line: Engine Size vs Price

The scatter plot with fitted regression line below isolates the relationship between engine size (cc) and price, which shows one of the clearest linear trends in the dataset. Training samples are shown in blue, test samples in orange, and the single-feature regression line is drawn in red. The positive slope of the regression line confirms the strong positive correlation between engine size and price. While the full multi-feature regression model uses all 24 features simultaneously, this univariate visualization effectively illustrates the core principle of linear regression: fitting the best-fitting straight line to explain the relationship between a single feature and the target variable.

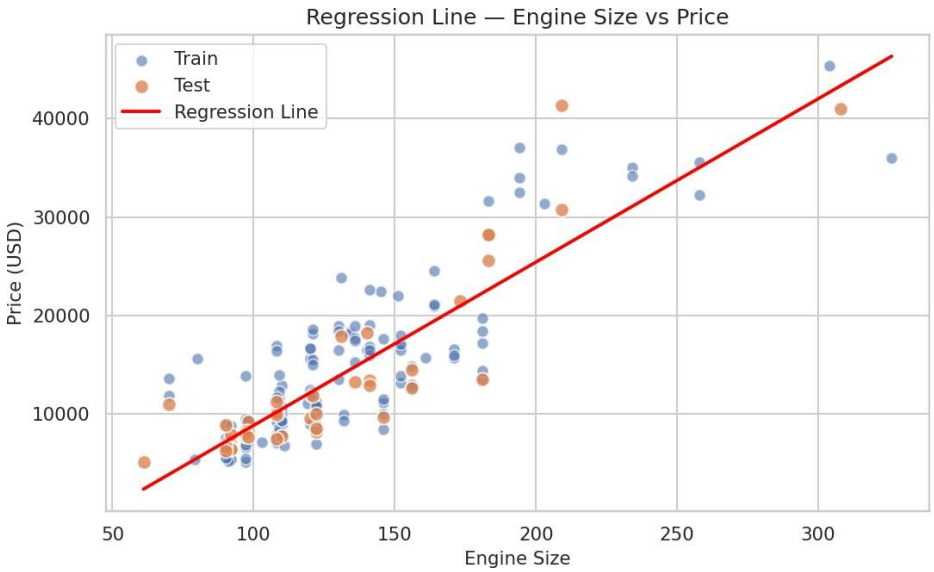


Figure 10: Regression Line — Engine Size vs Price

7.3 Limitations of the Model

While the Linear Regression model demonstrated strong predictive performance, several limitations should be acknowledged:

- Linearity assumption: Linear Regression assumes strictly linear relationships between features and price. Non-linear interactions, such as the interaction between brand reputation and engine size on price, cannot be captured.
- Multicollinearity: Several features such as wheelbase, carlength, curbweight, and enginesize are strongly intercorrelated. Multicollinearity inflates coefficient variance and can make individual coefficient estimates unstable.
- Heteroscedasticity: The residual plot reveals wider prediction errors at higher price points, violating the homoscedasticity assumption of OLS regression. A log-transformation of the target variable could partially address this.
- Dataset size: With only 205 records, the model has limited training data. Some car brands and body types have very few representatives, reducing confidence in predictions for those categories.
- Missing external factors: Vehicle condition, accident history, regional market demand, and economic conditions all influence car prices but are absent from the dataset.

Chapter 8 — Conclusion & Future Scope

8.1 Conclusion

This project successfully developed an end-to-end car price prediction system using Multiple Linear Regression applied to the CarPrice dataset. The complete machine learning pipeline was implemented: data ingestion, exploratory data analysis, data cleaning and preprocessing, label encoding, feature-label separation, train-test split, model training, evaluation, visualization, serialization, and real-time inference.

The preprocessing stage addressed the non-trivial challenge of extracting meaningful features from the raw dataset: the CarName column was parsed to extract manufacturer brands, known typos were corrected, and all categorical attributes were encoded using scikit-learn's LabelEncoder. The cleaned dataset contained 205 records and 24 features ready for modeling.

EDA through 10 distinct visualizations provided deep insights into the dataset: engine-related features (enginesize, horsepower, curbweight) are the strongest predictors of higher car prices, while fuel efficiency metrics (citympg, highwaympg) are inversely correlated with price. Brand identity captures meaningful pricing variance, with luxury brands commanding significantly higher average prices than economy brands.

The trained Linear Regression model achieved a Test R^2 of 0.8459, explaining 84.6% of price variance on unseen data. The Test RMSE of $3,487.68 is approximately 8.7% of the total price range, confirming strong predictive accuracy across the mainstream price segment. The regression equation with an intercept of -$74,838.36 and 24 feature-specific coefficients provides a fully interpretable mathematical model of car pricing.

The model was saved as car_price_model.pkl using joblib, enabling efficient real-time deployment. A demonstration prediction for a mid-range sedan yielded $12,066.29, which is consistent with the expected market price for that vehicle profile, validating the model's practical utility.

One of the principal advantages of Linear Regression in this context is its interpretability. The regression coefficients directly quantify the contribution of each vehicle attribute to the predicted price. For example, each unit increase in encoded enginelocation (front to rear) corresponds to a price increase of $11,651 — a reflection of the fact that rear-engine cars in this dataset are predominantly high-performance vehicles. These insights can be directly acted upon by automotive analysts and pricing specialists.

8.2 Future Scope

Several avenues exist to improve and extend this car price prediction system in the future:

- Advanced algorithms: Implementing ensemble methods such as Random Forest Regression, Gradient Boosting (XGBoost, LightGBM), or Neural Networks could capture non-linear feature interactions and improve prediction accuracy, particularly for premium-priced vehicles.
- Log transformation of target: Applying a log transformation to the price column before training would normalize the skewed distribution, reduce heteroscedasticity, and likely improve model performance at both ends of the price spectrum.
- Larger datasets: Augmenting the dataset with additional vehicle records from sources like Kaggle or automotive databases would reduce overfitting and improve generalization, especially for underrepresented categories.
- Web application deployment: The serialized joblib model could be integrated into a Flask or FastAPI web application, enabling users to enter vehicle specifications and receive instant price estimates via a browser interface.

- Feature engineering: Creating interaction features (e.g., horsepower-to-weight ratio, enginesize-per-cylinder) or polynomial features could help capture non-linear pricing effects within a linear regression framework.
- Real-time data integration: Incorporating live market data feeds from automotive listing platforms would allow the model to adapt to current market conditions and regional pricing differences.
- Deep Learning: A Multi-Layer Perceptron (MLP) regression model could be trained on a larger dataset to capture complex, non-linear relationships between features and price that linear regression cannot represent.

Car Price Prediction: A Machine Learning Analysis

Summarising the methodology and performance of a Linear Regression model to predict car resale values using technical specifications.

DATASET OVERVIEW

205 UNIQUE VEHICLES

The final cleaned dataset consists of 205 car samples across various brands and body types.

25 PREDICTIVE FEATURES

The model analyses 25 distinct attributes including dimensions, engine specifications, and fuel systems.

PRIMARY MARKET SEGMENTS

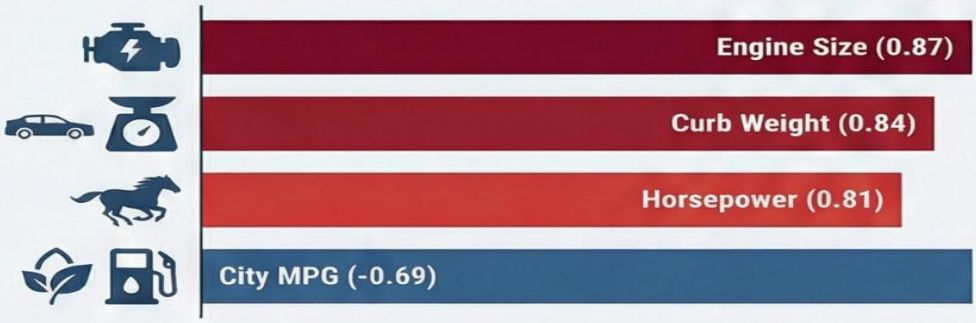
Dominated by petrol-fuelled cars (30.2%) and Front-Wheel Drive (FWD) configurations (58.5%).

Most Frequent Car Body Types

Body Type	Frequency
Sedan	96
Hatchback	70
Wagon	25
Hardtop	8
Convertible	6

KEY CORRELATIONS

High Positive Correlation — High Negative Correlation



Engine Size (0.87)
Strungest positive relationship; larger engines drastically increase predicted value.

Curb Weight (0.84)
Secondary major predictor; high positive correlation with market value.

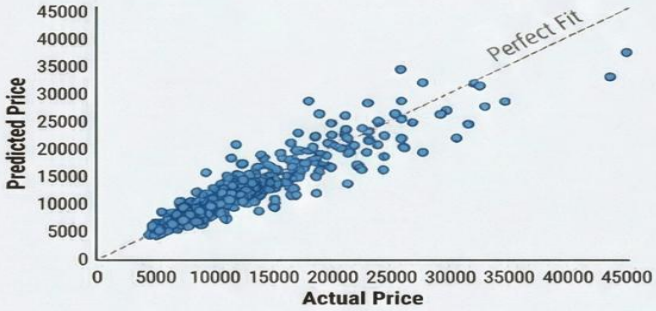
Horsepower (0.81)
Performance boosts price.

City MPG (-0.69)
Negatively correlated; efficiency metrics often contradict high value.

MODEL PERFORMANCE & TOP DRIVERS

MODEL PERFORMANCE (ACTUAL VS PREDICTED)

The model successfully explains 84.6% of the variance in car prices.



TEST R^2 SCORE: 0.8459

RMSE: $3,487
Average devistion from actual values.

MAE: $2,090
Average magnitude of errors.

TOP PRICE DRIVERS (COEFFICIENTS)

Engine Location (Rear)
Drastically increases predicted value.

Fuel Type (Diesel/Gas)
Critical factor in regression.

Engine Stroke Length
Significant negative impact, acting as downward pressure on price.

References

The following resources were referenced in the development and documentation of this project:

- CarPrice Dataset — Kaggle (https://www.kaggle.com/datasets/hellbuoy/car-price-prediction)
- scikit-learn Documentation — LinearRegression, LabelEncoder, train_test_split, metrics — https://scikit-learn.org
- pandas Documentation — DataFrame operations, groupby, describe — https://pandas.pydata.org
- matplotlib & seaborn Documentation — Visualization APIs — https://matplotlib.org / https://seaborn.pydata.org
- joblib Documentation — Model serialization and loading — https://joblib.readthedocs.io
- Géron, A. (2022). Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow. O'Reilly Media.
- James, G. et al. (2021). An Introduction to Statistical Learning. Springer.
- Online resources including Python documentation, Stack Overflow, and Anthropic Claude were used to understand implementation details and help structure the analysis and report.

Also Marking my Practical Works upon this project looking into the class resources and the references –

- AryashreeS/CarPredictionModel: Config files for my GitHub profile.